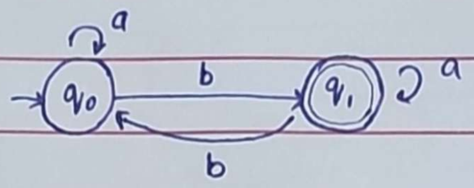
DATE:

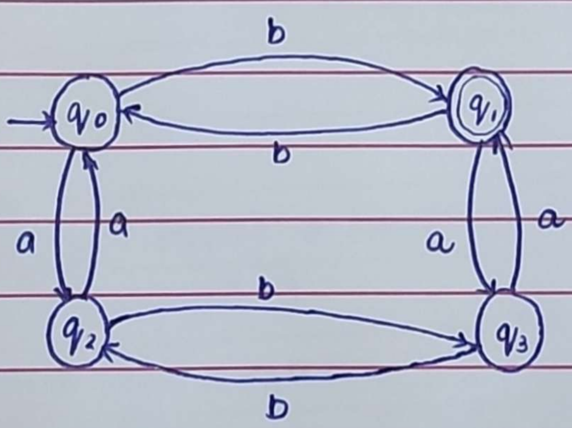
#01 DFA over alphabet {a, b} that recognizes all strings containing an odd number of b's.



Transition Table

	Current State	a	b
start	q_0	q_0	q_1
final	q_1	q_1	q_0

#02 DFA that recognizes all strings containing an even number of a's and odd b's.

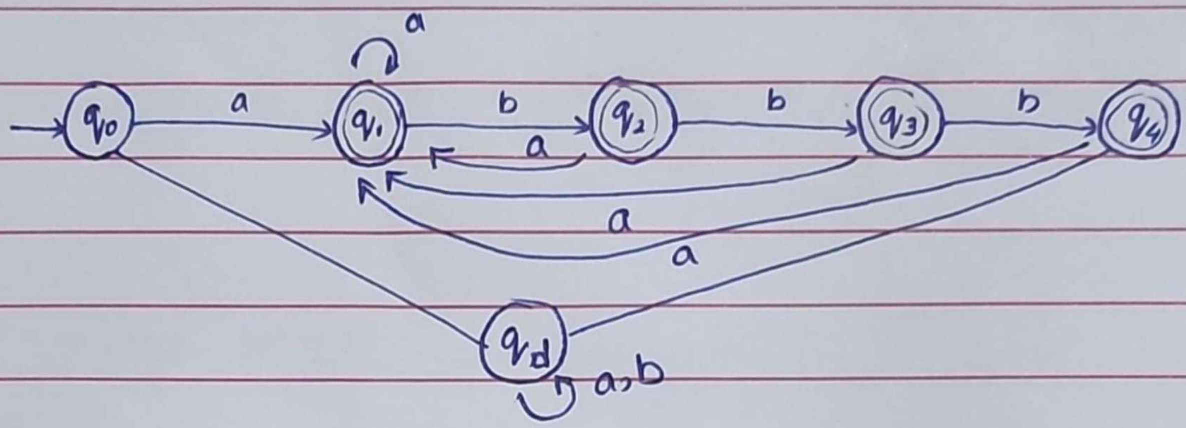


Transition Table

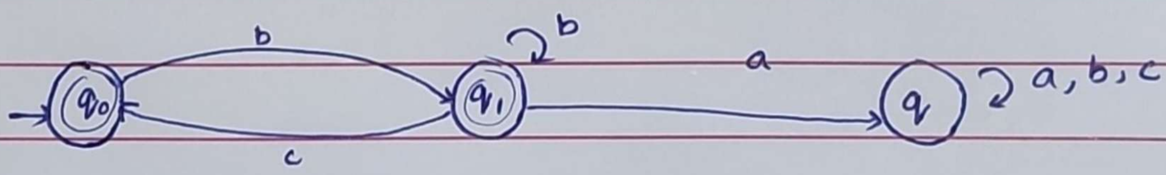
	current	a	b
start	q_0	q_2	q_1
final	q_1	q_3	q_0
	q_2	q_0	q_3
	q_3	q_1	q_2

DATE: ______

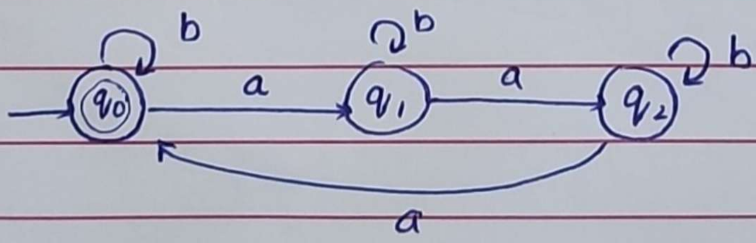
03 DFA for ~~(a)~~ (a|ab|abb|abbb)*



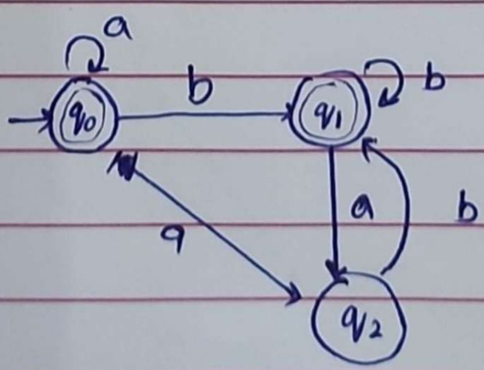
04 DFA over {a, b, c} accepts all strings not containing substring "ba"



05 DFA over {a, b} accepts all strings where number of a's is a multiple of 3.



06 DFA that accepts all over {a, b} that begin or end with the substring "ba"



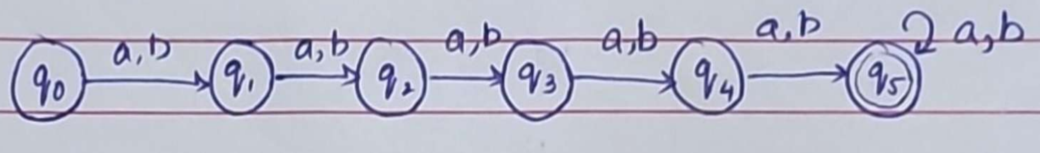
Transition Table

state	a	b	accept
q_0	q_2	q_1	YES
q_1	q_3	q_1	YE
q_2	q_2	q_1	YES
q_3	q_3	q_1	YES

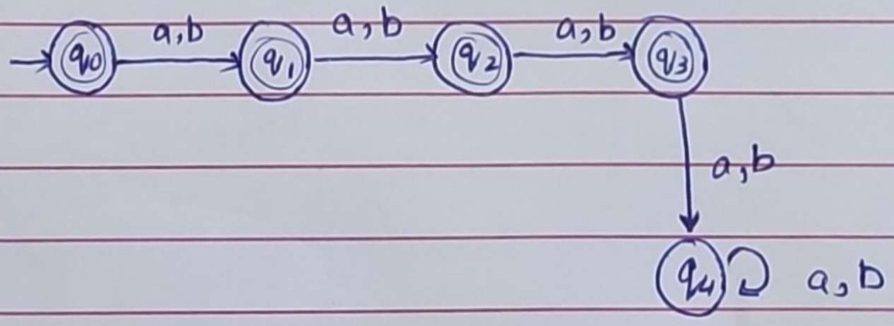
DATE:

#08

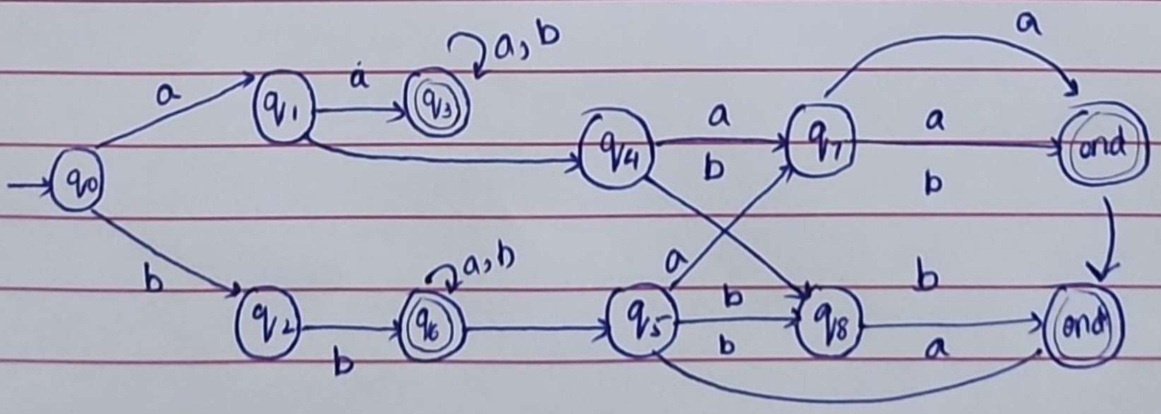
i. DFA that accepts all strings having more than four letters.



ii. DFA that accepts all strings having fewer than four letters.



#07



state	a	b	accept
start → q0	q1		
q1	q3		
q~~1~~2	q5		
q~~4~~3	q3		
q~~6~~4	q6		
q~~6~~5	q7		
q~~7~~6	q7		
q8 7	endaa		
q8	q7		

Transition Table

	State	a	b	accept
start →	q_0	q_1	q_2	NO
	q_1	q_3	q_4	NO
	q_2	q_5	q_6	NO
	q_3	q_3	q_3	YES
	q_4	q_6	q_6	YES
	q_5	q_7	q_8	NO
	q_6	q_7	q_8	NO
	q_7	end aa	q_8	NO
	q_8	q_7	end-bb	NO

$\left\{ \begin{array}{l} \text{end aa} \rightarrow q_7 \\ \text{end bb} \rightarrow q_7 \end{array} \right\}$ yes